

IGCSE Mathematics 0580 — Every Formula You Need (Extended)

Cambridge IGCSE 0580, exams 2025-2027. Mensuration formulae are given in the exam; everything here beyond that must be memorised.

NUMBER

Percentage change	$= (\text{new} - \text{original}) / \text{original} \times 100$
Reverse percentage	$\text{original} = \text{new} / (1 + \text{rate})$ e.g. after 20% increase: divide by 1.2
Compound interest	$\text{final} = P (1 + r/100)^n$
Simple interest	$I = P r t / 100$
Speed / density / pressure	speed = d / t ; density = m / V ; pressure = F / A
Index laws	$a^m \times a^n = a^{(m+n)}$; $a^m / a^n = a^{(m-n)}$; $(a^m)^n = a^{(mn)}$
Negative & fractional	$a^{-n} = 1/a^n$; $a^{(1/2)} = \sqrt{a}$; $a^{(m/n)} = (\text{n-th root of } a)^m$
Standard form	$A \times 10^n$, where $1 \leq A < 10$

ALGEBRA

Quadratic formula	$x = (-b \pm \sqrt{b^2 - 4ac}) / 2a$
Difference of two squares	$a^2 - b^2 = (a - b)(a + b)$
Expanding $(a+b)(c+d)$	$= ac + ad + bc + bd$
nth term (linear)	$T_n = an + b$ (a = common difference)
nth term (quadratic)	$T_n = an^2 + bn + c$ (find a, b, c from terms)
Simultaneous equations	Elimination or substitution; check answer in BOTH equations
Composite functions	$fg(x) = f(g(x))$ — apply g FIRST
Inverse function	$f^{-1}(x)$: swap x and y in $y = f(x)$, then make y the subject

GEOMETRY & MENSURATION

Polygon angles	interior sum = $(n - 2) \times 180$; exterior = $360 / n$
Pythagoras	$a^2 + b^2 = c^2$ (right angle only)
Similar shapes	lengths: $x k$; areas: $x k^2$; volumes: $x k^3$
Sector arc length	$s = (\theta / 360) \times 2 \pi r$
Sector area	$A = (\theta / 360) \times \pi r^2$

CIRCLE THEOREMS — learn the reasons word-for-word

Angle at centre	angle at centre = $2 \times$ angle at circumference (same arc)
Same segment	angles in the same segment are equal
Semicircle	angle in a semicircle = 90 degrees
Cyclic quadrilateral	opposite angles sum to 180 degrees
Tangent	tangent meets radius at 90 degrees
Alternate segment	angle between tangent and chord = angle in alternate segment

TRIGONOMETRY

SOHCAHTOA	$\sin = \text{opp/hyp}$; $\cos = \text{adj/hyp}$; $\tan = \text{opp/adj}$
Sine rule	$a / \sin A = b / \sin B = c / \sin C$
Cosine rule	$c^2 = a^2 + b^2 - 2ab \cos C$; $\cos C = (a^2 + b^2 - c^2) / 2ab$
Triangle area	$A = (1/2) a b \sin C$
Exact values	$\sin 30 = 1/2$; $\cos 60 = 1/2$; $\tan 45 = 1$; $\sin 60 = \sqrt{3}/2$; $\tan 30 = 1/\sqrt{3}$

STATISTICS & PROBABILITY

Mean (list)	mean = sum of values / number of values
Mean (table)	mean = $\sum(fx) / \sum(f)$
Moving averages	plot successive group means; comment on trend
Probability basics	$P(\text{event}) = \text{favourable} / \text{total}$; all outcomes sum to 1
Independent events	$P(A \text{ and } B) = P(A) \times P(B)$
Conditional (tree)	$P(B A) = P(A \text{ and } B) / P(A)$ — denominator changes!